

COLX 585: Membership Attack in VLMs Project: Milestone 4

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Abstract

Membership inference attack (MIA) is a significant area of AI safety research. With the development of larger and larger models, including vision language models (VLMs), new MIA methods, particularly in multimodal contexts, are becoming increasingly necessary. In this milestone, we set out to understand the current research in this field by summarizing four papers, and begin our own methods through a baseline model and data exploration. We developed a simple baseline for membership inference using the feature extraction and classification pipeline in milestone4. We then trained two simple classifiers, Logistic Regression and XGBoost, and additionally constructed an ensemble by averaging their predicted probabilities. The best-performing model was used to generate test-set membership predictions in Kaggle submission format, and the AUC was 0.7793 and $\text{TPR@FPR=0.1} = 0.2833$.

1 Introduction

Membership inference attack (MIA) is a significant area of AI safety research. With the development of larger and larger models, including vision language models (VLMs), new MIA methods, particularly in multimodal contexts, are becoming increasingly necessary. In this milestone, we set out to understand the current research in this field by summarizing four papers, and begin our own methods through a baseline model and data exploration.

2 Background

Summary of related research papers:

2.1 Paper 1: "Tracing Training Footprints: A Calibration Approach for Membership Inference Attacks Against Multimodal Large Language Models" (Zheng et al., Findings 2025)

This paper focuses on detecting whether private, copyrighted or unauthorized data has been used to

train MLLMs. It employs a Membership-Inference Attack ("MIA") method to determine whether specific data examples were used in a training set. The paper notes that previous MIA methods focus only text, which perform poorly when image embeddings are present as they provide extensive information to the model that makes many tokens easy to predict regardless of membership.

The authors propose PC-MMIA, a perturbation-based calibration method that measures the difference in token log-probability before and after image perturbation. The differences are assigned as weights to the token's log-probability, reflecting the assumption that high-weight (high difference) tokens indicate the model saw that image-caption pair in training, and that the token is not just easy to predict.

Experiments are conducted on LLaVA-1.5-7b and CogVLM-17b models, chosen for being open-source and having different architectures, which supports reproducibility and experimental diversity. Datasets used are NoCaps (166,100 human-written captions for 15,100 images), Flickr30K (31,000 images with 5 human-written captions each), and PixMo (a diverse dataset with images and captions produced synthetically and by humans).

2.2 Paper 2: "Membership Inference Attacks against Large Vision-Language Models" (Li et al., NeurIPS 2024)

The paper sets out to address user concerns regarding the use of private information such as photographs and medical records in training VLLMs, which can leak training data. This paper focuses on single-modality, determining if an image was seen in training, without addressing the caption. The authors suggest that in reality, a user wanting to verify if their image was used to train a model would have access only to the image, and not the associated caption that was generated for it. The models used are LLaVA 1.5, MiniGPT-4 and LLaMA-Adapter

V2.

The paper proposes three main contributions. The first MIA benchmark specifically for VLLMs, called VL-MIA, a group of standardized datasets constructed from MS COCO images as members, with non-members sourced from Flickr and DALL-E for image tasks, and GPT-4 generated text as non-members for text tasks. Second, a cross-modal pipeline where the VLM generates a caption for an input image and instruction, then the image, instruction and caption are given to the VLM together, so that text-based MIA methods can also be applied to image MIA. Third, MaxRényi-K% (and the target-based version ModRényi), which measures the model's confidence in predicting each token, where a low score indicates membership as the model is more confident in its predictions.

2.3 Paper 3: "Membership Inference Attacks Against Vision-Language Models" (Hu et al., USENIX Security 2025)

In this paper, Hu et al. introduce new methods to perform membership inference attacks on Vision Language Models (VLMs), since previous MIA attack methods are not effective on very large models like VLMs. The paper evaluates its attacks on vision-language models including LLaVA and MiniGPT-4, using different LLM backbones such as Vicuna (7B, 13B) and Llama-2 (7B, 13B). Since large models only provide "black box" access, new methods are needed to distinguish whether or not the model has seen particular data in training. Starting with the understanding that VLMs / LLMs use the temperature parameter, which is applied in the softmax function, to control how random or confident the generated output is. Low temperature means that the model is going to generate very confident output and be closer to a deterministic response. High temperature means that the model is going to generate more random output. The techniques described in this paper exploit the fact that members are more sensitive to temperature parameter changes than non-members.

Using members' temperature sensitivity, four new methods for attack are: shadow model inference, reference inference, target-only inference, and image-only inference. In shadow model inference, a shadow model is trained to learn the patterns of member vs non-member and train a classifier. While this method is the most powerful, it is also the least realistic, due to the amount of data and compute required. Reference inference

determines if a dataset was used in training by comparing its output behavior to a known reference set (non-members) and checking if their statistics are significantly different. The third method, target-only inference categorizes membership by querying the same set of data at different temperatures and checking if the model's outputs change significantly, since training data is more sensitive to temperature shifts. Lastly, image-only inference determines membership by using images to query the model repeatedly and measure how consistent the generated descriptions are, with higher consistency indicating that an image is likely training data.

2.4 Paper 4: "Practical Membership Inference Attacks Against Large-Scale Multi-Modal Models: A Pilot Study" (Ko et al., 2023)

This paper proposes three methods of membership inference attacks against large multi-modal models, like CLIP. The paper discusses attacks on multiple CLIP (OpenCLIP) models, including Vision Transformer variants (ViT-B/32, ViT-B/16, ViT-L/14) and ResNet-based models (RN50, RN101) across different datasets. CLIP (Contrastive Language-Image Pretraining) was developed by Open AI as a large model where images and text share the same embedding space. Because the only access to a model is often "black box" access, where users can only enter queries and observe results, the three methods put forth by this paper do not require any special access to the model. The paper includes three attack methods: Cosine Similarity Attack (CSA), Augmentation-Enhanced Attack (AEA), and Weakly Supervised Attack (WSA). CSA uses the knowledge that image and text pairs have high cosine similarity. Non-matching image and text pairs have low cosine similarity. The attack is to input a pair of image and text into CLIP. Then, cosine similarity is computed between embeddings and compared to a threshold. Training data pairs sit measurably "closer" in embedding space than non-members. AEA uses the approach of applying transformations, such as crop, rotate, resize, to the image and then measuring how similarity changes, noting how training samples are more sensitive to changes. In WSA, the attacker uses real non-member data, data created after the model release date to estimate the distribution of similarity for non-members. Then it identifies data that are likely to be members. Using this information, an attack classifier is trained.

In addition to discussing the three attacks, this paper explores four defenses, including regularization, data augmentation, noise injection, and differential privacy. The first defense uses L2 regularization to add a penalty during training to reduce overfitting. While this may help limit membership data leakage, it also hurts performance. In data augmentation, transformations are applied to training data to improve generalization of data. In feature perturbation or noise injection, Gaussian noise is added to model outputs in order to reduce membership signals. This may significantly reduce the quality of the model. Lastly, differential privacy adds noise to change individual data points, but is difficult to implement efficiently for large models and can cause problems in training.

3 Data

The data for this task include 6000 training samples, 1200 validation samples, and 6000 test samples, which all contain pairs of images and descriptions. Each example in the training and validation splits has the following metadata: id, image, is_member, type, and text. Type indicates whether the target VLM has seen both the text and image, just the image, just the text, or neither the image or the text. Since this task is multimodal, these tags allow us to analyze the VLM behaviour in more detail: an instance is a member if the target VLM has seen both the image and text; all other types are labeled as 0. The test split does not have any membership information.

In total, the train set has 1500 members and 4500 non-members; the validation set has 300 members and 900 non-members. There is an even number of types across the training and validation data, with 1500 of each type in train and 300 in validation. The texts average around 225-230 characters across all three splits, though there is some variation in text length, compared to image size, which are all 512x512 pixels. After plotting the text length distribution by train, validation, and test splits, we found that all three splits have similar distributions of text length, right-skewed with a peak at around 150-200 characters and a long tail out to over 600 characters. The consistency in text length and image size shows that there will be no bias in length that could influence the MIA classifier. See the table below for more details.

Table 1: Dataset Statistics

	Train	Validation	Test
Num Samples	6000	1200	6000
Mean Text Length	229.5	224.43	228.75
Max Text Length	776	698	1060
Min Text Length	69	77	62
Median Text Length	214	208	214
Members (1)	1500	300	0
Non-Members (0)	4500	900	0
Member Ratio	0.25	0.25	0.0
Mean Img Width	512	512	512
Mean Img Height	512	512	512
Min Img Size	512x512	512x512	512x512
Max Img Size	512x512	512x512	512x512
Type: unseen_both	1500	300	–
Type: seen_img_unseen_txt	1500	300	–
Type: seen_both	1500	300	–
Type: unseen_img_seen_txt	1500	300	–
Type: None	–	–	6000

4 Week 1 Baseline

We developed a simple baseline for membership inference using the feature extraction and classification pipeline in milestone4. The baseline consists of two stages: feature extraction from the official dataset and binary classification over the resulting feature representations.

In the first stage, we loaded the official dataset splits (train, validation, and test) using the Hugging Face dataset loader. Each image-text example was processed by both the fine-tuned target vision-language model and a base reference model. From the resulting model outputs, we extracted a set of numerical features that characterize the model’s behavior on each example. These features include loss-based statistics, perplexity, token-probability statistics, entropy-related quantities, rank-based measures, and comparative features derived from the difference between the target model and the reference model. The extracted features were then stored as tabular data in train.csv, val.csv, and test.csv.

In the second stage, we formulated membership inference as a standard binary classification task over the extracted feature tables. Each row corresponds to a single example, and the target label indicates whether the example is a member (1) or a non-member (0). Prior to training, the pipeline automatically selected the numeric feature columns, handled missing and infinite values, and applied

feature standardization when appropriate. We then trained two simple classifiers, Logistic Regression and XGBoost, and additionally constructed a simple ensemble by averaging their predicted probabilities. Model selection was performed on the validation split, and the best-performing model was used to generate test-set membership predictions in Kaggle submission format.

The motivation for this baseline was to establish a method that is straightforward, reproducible, and interpretable. Rather than introducing a highly engineered attack at the initial stage, we adopted simple classifiers over extracted model-behavior features in order to obtain a clear and well-defined reference point for subsequent improvements.

The feature extraction stage was designed for a Colab-style environment and used GPU acceleration when available, since inference with vision-language models constitutes the main computational cost. Overall, this baseline provides a practical initial system that transforms official dataset examples into feature vectors, applies simple binary classifiers, evaluates them on validation data, and produces a submission file for the test set.

The XGBoost classifier had the best performance with $AUC = 0.7793$ and $TPR@FPR=0.1 = 0.2833$. Our Kaggle submission scored $AUC = 0.76850$.

Table 2: Model Performance Comparison

Method	AUC	TPR@FPR=0.1
Logistic Regression	0.7760	0.2800
XGBoost	0.7793	0.2833
Ensemble (Average)	0.7790	0.2600

In addition, we collected data on the importance of the features we used in the baseline model. The most important features were caption_loss_diff and perplexity_ratio.

References

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- [2] Myeongseob Ko, Ming Jin, Chenguang Wang, and Ruoxi Jia. 2023. Practical Membership Inference Attacks Against Large-Scale Multi-Modal Models: A Pilot Study. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4871–4881. [cite: 1411, 1412]

Table 3: Top 10 Individual Feature AUCs

Feature Name	Individual AUC
caption_loss_diff	0.7299
perplexity_ratio	0.7299
ref_min_k10_prob	0.6879
ref_caption_loss	0.6813
ref_caption_perplexity	0.6813
loss_ratio	0.6793
ref_min_k20_prob	0.6730
ref_min_token_prob	0.6620
mean_prob_diff	0.6175
mean_entropy_diff	0.6109

- [3] Zhan Li, Yongtao Wu, Yihang Chen, Francesco Tonin, Elias Abad Rocamora, and Volkan Cevher. 2024. Membership Inference Attacks against Large Vision-Language Models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 37, pages 98645–98674.
- [4] Xiaofan Zheng, Huixuan Zhang, and Xiaojun Wan. 2025. Tracing Training Footprints: A Calibration Approach for Membership Inference Attacks Against Multimodal Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 17179–17191.

5 Contributions

5.1 Milestone 4

Christina: EDA, data section, introduction : 0.25
Jennifer: Baseline version 1, papers 3 and 4: 0.25
Rachelle: Papers 1 and 2, editing: 0.25
Yucai: Baseline version 2, baseline model discussion: 0.25